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ER

Entity Relationship Model (ERM)

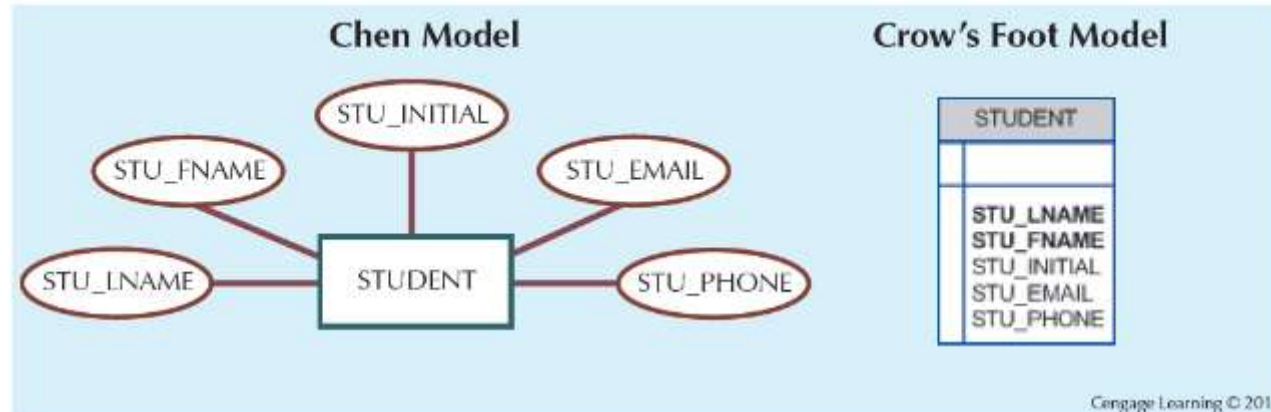
- Basis of an entity relationship diagram (ERD)
- ERD depicts the:
 - Conceptual database as viewed by end user
 - Database's main components
 - Entities
 - Attributes
 - Relationships
- Entity - Refers to the entity set and not to a single entity occurrence

Attributes

- Characteristics of entities
- **Required attribute:** Must have a value, cannot be left empty
- **Optional attribute:** Does not require a value, can be left empty
- **Domain** - Set of possible values for a given attribute
- **Identifiers:** One or more attributes that uniquely identify each entity instance

An identifier is also called a KEY, or PRIMARY KEY - **this is one of the 'key' concepts in all of database theory!!** We'll talk much more about keys later.

Figure 4.1 - The Attributes of the Student Entity: Chen and Crow's Foot



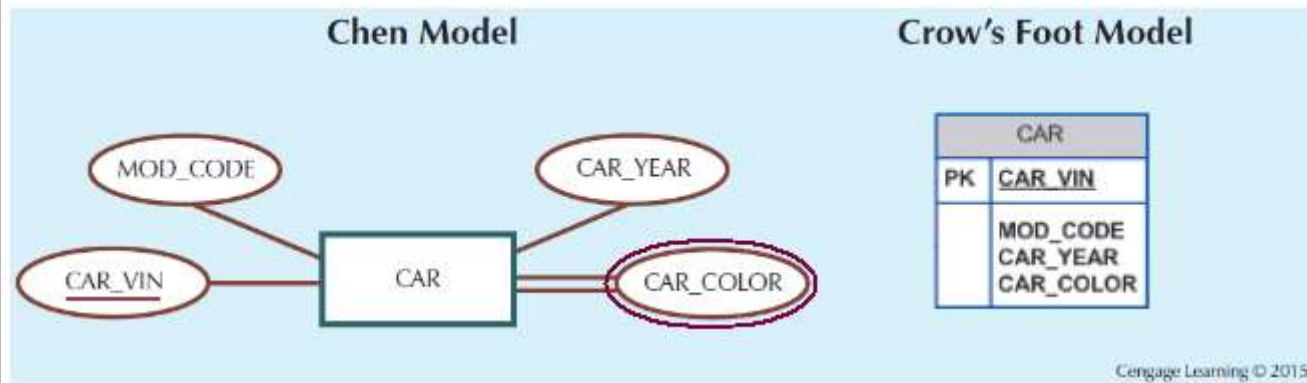
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Attributes

- **Composite identifier:** Primary key composed of more than one attribute
- **~~Compound~~ Composite attribute:** Attribute that can be subdivided to yield additional attributes
- **Simple attribute:** Attribute that cannot be subdivided
- **Single-valued attribute:** Attribute that has only a single value
- **Multivalued attributes:** Attributes that have many values

FYI - here is a page on the various types of attributes.

Figure 4.3 - A Multivalued Attribute in an Entity



In Crow's Foot notation, 'bold' attributes are 'required' (can't be null).

Attributes

- **Multivalued attributes:** Attributes that have many values and require creating:
 - Several new attributes, one for each component of the original multivalued attribute
 - A new entity composed of the original multivalued attribute's components
- **Derived attribute:** Attribute whose value is calculated from other attributes
 - Derived using an algorithm

Figure 4.6 - Depiction of a Derived Attribute

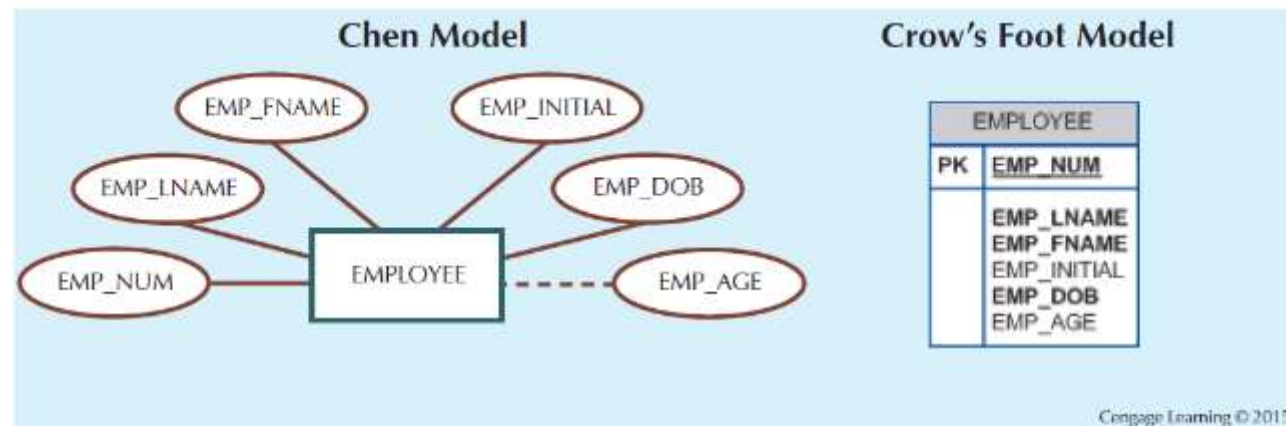


Table 4.2 - Advantages and Disadvantages of Storing Derived Attributes

	STORED	NOT STORED
Advantage	Saves CPU processing cycles Saves data access time Data value is readily available Can be used to keep track of historical data	Saves storage space Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	Uses CPU processing cycles Increases data access time Adds coding complexity to queries

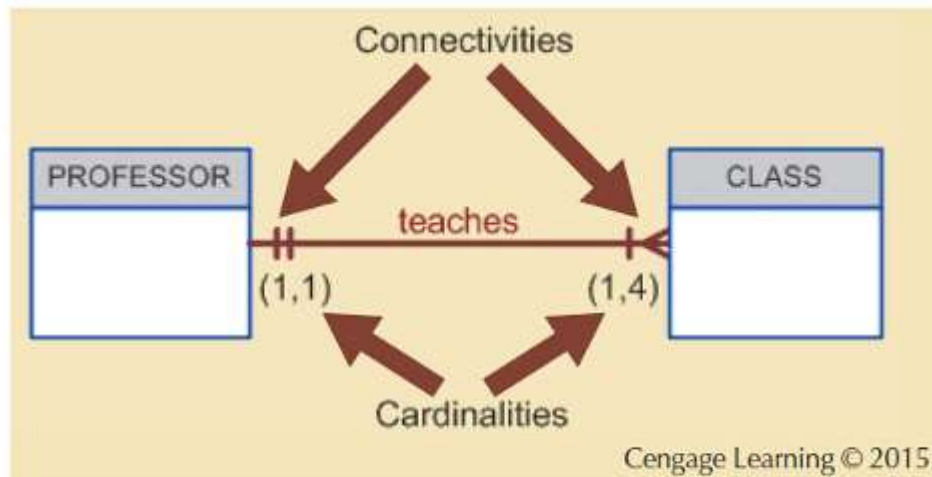
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Relationships

- Association between entities that always operate in both directions
- **Participants:** Entities that participate in a relationship
- **Connectivity:** Describes the relationship classification
- **Cardinality:** Expresses the minimum and maximum number of entity occurrences associated with one occurrence of related entity

Connectivity vs cardinality

Figure 4.7 - Connectivity and Cardinality in an ERD



Connectivity: 1:1, 1:M or M:N (three diff ways by which two entities are related).

Cardinality: (min,max) for 1:1, 1:M or M:N (eg. 1:1 can have (1,0) as its cardinality, 1:M can have (0,4) as its cardinality). Sometimes, min is called 'modality' (and max is cardinality). The 'inside' symbols denotes min, and the outside ones, max.

Confusingly, the # rows in a table is ALSO called table's cardinality (and, # of columns is called the table's degree).

Also confusingly, 1:1, 1:M, M:N are called 'cardinality ratios'!

'Can I exist apart from you?'

Existence Dependence

Existence dependence

- Entity exists in the database only when it is associated with another related entity occurrence

Existence independence

- Entity exists apart from all of its related entities
- Referred to as a **strong entity** or **regular entity**

Existence independence implies a strong entity; but, existence dependence (alone, ie. by itself) does NOT imply a weak entity (there needs to be one more condition, based on 'relationship strength', for it to become 'weak').

In other words, **we need to look at where the FK in the dependent entity is located.**

Existence dependence

An entity B is "existent dependent" on another entity A, if, a row in B can only exist when its FK is NOT NULL, ie. a corresponding entry exists in A.

Eg. if A is EMPLOYEE and B is DEPENDENT, a dependent (eg. child) in B can only exist if there is a corresponding employee (eg. Dad) in A. THIS ALONE DOES NOT MAKE 'B' A WEAK ENTITY!

Weak vs strong relationship

Again, it's all about the FK [WHERE it goes], in the dependent entity!

Relationship Strength

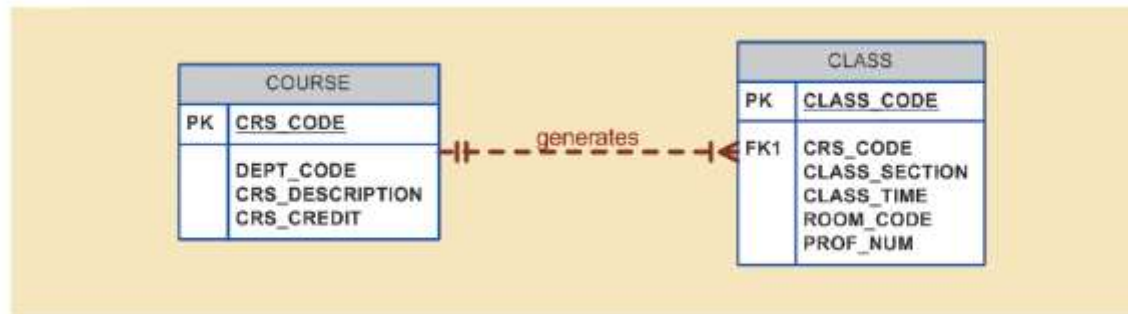
Weak (non-identifying) relationship

- Primary key of the related entity does not contain a primary key component of the parent entity

Strong (identifying) relationships

- Primary key of the related entity contains a primary key component of the parent entity

Figure 4.8 - A Weak (Non-Identifying) Relationship between COURSE and CLASS

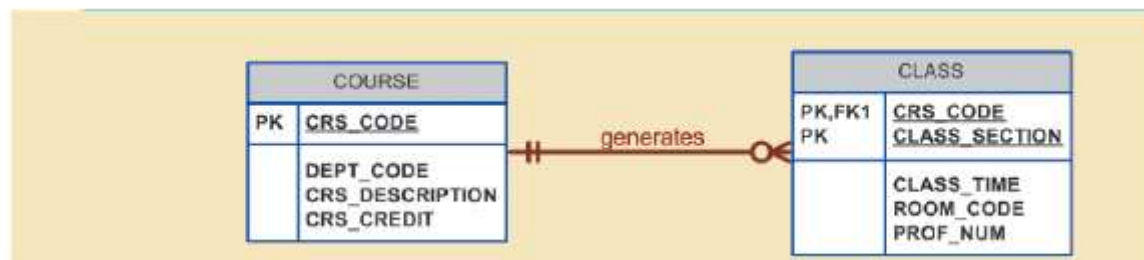


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So, here, CLASS is **not** a weak entity.

Strong ("common PK") course-class relation

Figure 4.9 - A Strong (Identifying) Relationship between COURSE and CLASS



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CLASS is now a weak entity (because: it is existence dependent, AND has a strong relationship).

Weak entity [two conditions]

Weak Entity

- Conditions
 - Existence-dependent
 - Has a primary key that is partially or totally derived from parent entity in the relationship
- Database designer determines whether an entity is weak based on business rules

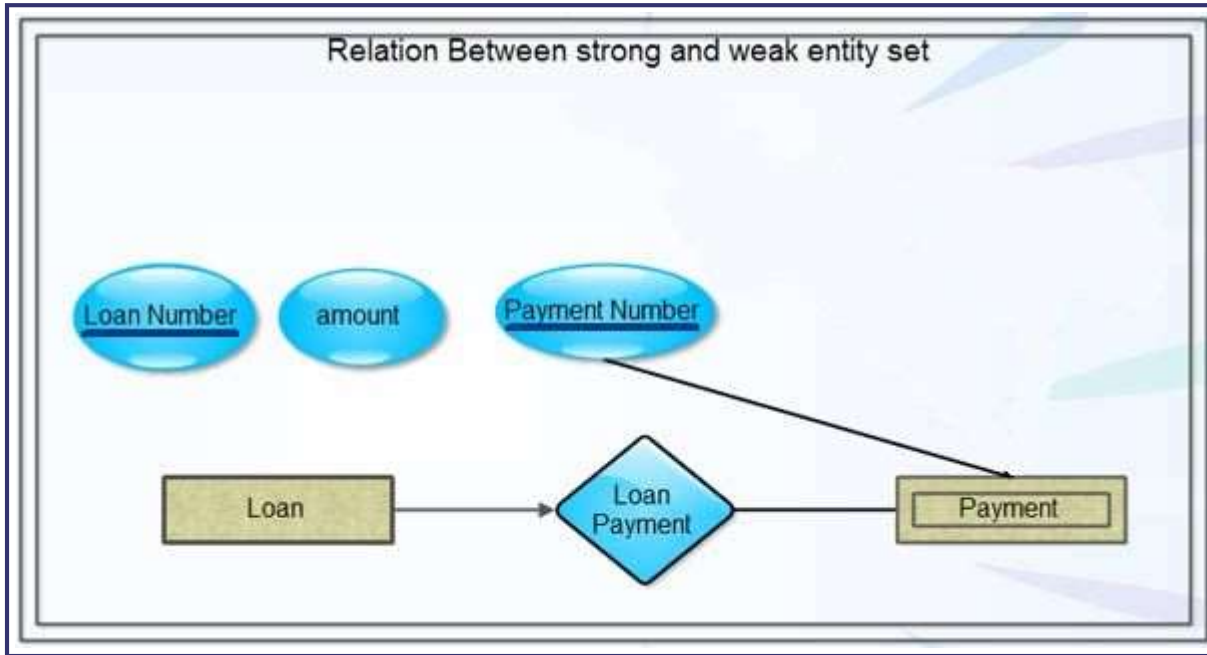
A weak entity needs to satisfy two conditions: existence dependence, strong (identifying/owning) relationship with a parent.

Note that a weak entity implies existence dependence, but existence dependence does not imply a weak entity!

Note too that a weak entity implies a strong ("owning" or "identifying") relationship.

Removing the controlling (owning) entity's key from a weak entity's PK will result in ****duplicates**** for remaining PK(s) - THAT is what makes it 'weak'.

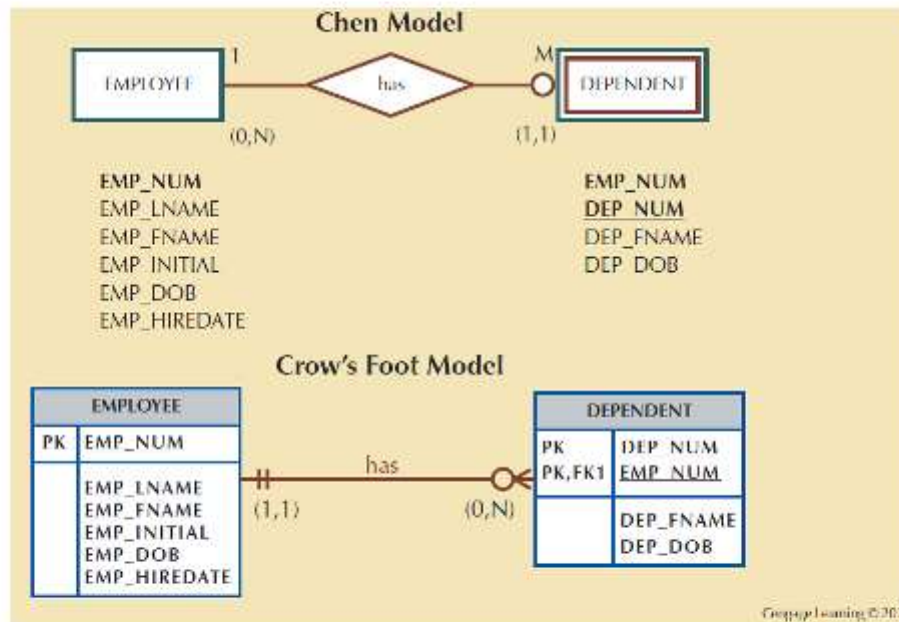
Weak entity - example



Payment cannot exist independent of Loan, AND needs Loan's key to be part of its own key, so it is a weak entity.

Weak entity

Figure 4.10 - A Weak Entity in an ERD



Weak entity

Figure 4.11 - A Weak Entity

Table name: EMPLOYEE

Database name: Ch04_ShortCo

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
1001	Callifante	Jeanine	J	12-Mar-64	25-May-97
1002	Smithson	William	K	23-Nov-70	28-May-97
1003	Washington	Herman	H	15-Aug-68	28-May-97
1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
1005	Johnson	Melanie		28-Sep-66	20-Dec-98
1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

Table name: DEPENDENT

EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
1001	1	Annelise	05-Dec-97
1001	2	Jorge	30-Sep-02
1003	1	Suzanne	25-Jan-04
1006	1	Carlos	25-May-01
1008	1	Michael	19-Feb-95
1008	2	George	27-Jun-98
1008	3	Katherine	18-Aug-03

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Relationship Participation

Optional participation

- One entity occurrence does not require a corresponding entity occurrence in a particular relationship

Mandatory participation

- One entity occurrence requires a corresponding entity occurrence in a particular relationship

Table 4.3 - Crow's Foot Symbols

CROW'S FOOT SYMBOLS	CARDINALITY	COMMENT
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

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Figure 4.13 - CLASS is Optional to
COURSE

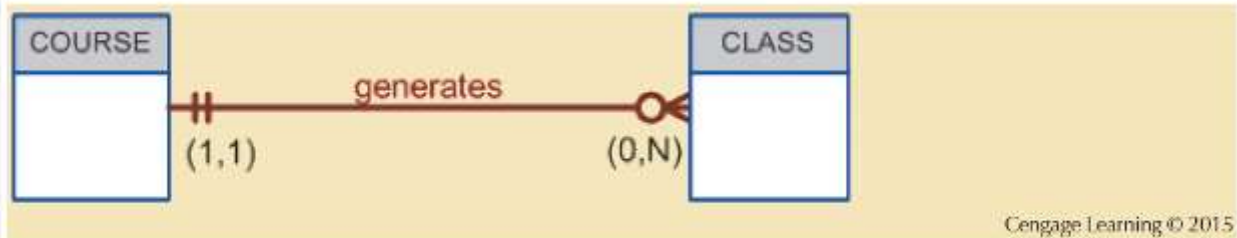


Figure 4.14 - COURSE and CLASS in a Mandatory Relationship



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Relationship Degree

- Indicates the number of entities or participants associated with a relationship
- **Unary relationship**: Association is maintained within a single entity
 - **Recursive relationship**: Relationship exists between occurrences of the same entity set
- **Binary relationship**: Two entities are associated
- **Ternary relationship**: Three entities are associated

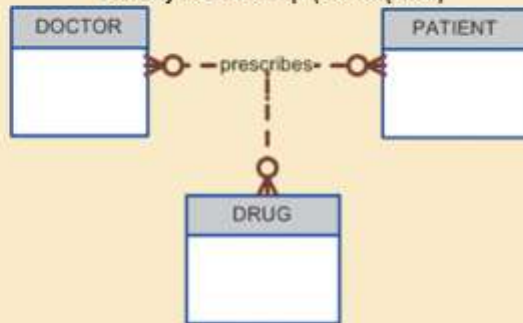
Unary relationship



Binary relationship



Ternary relationship (Conceptual)



Ternary relationship (Logical)

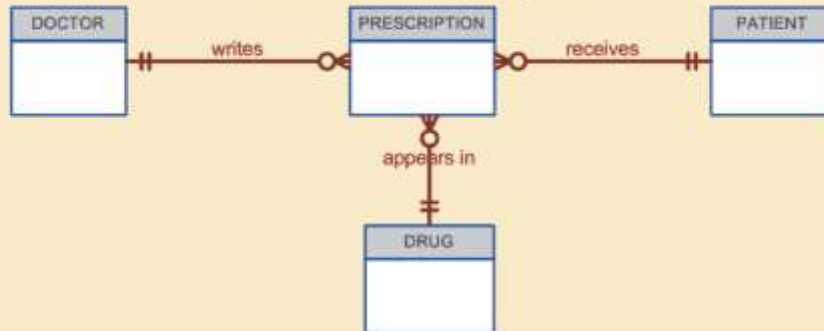


Figure 4.17 - An ER Representation of Recursive Relationships



Bridge entities

Associative Entities

- Also known as composite or bridge entities
- Used to represent an M:N relationship between two or more entities
- Is in a 1:M relationship with the parent entities
 - Composed of the primary key attributes of each parent entity
- May also contain additional attributes that play no role in connective process

Figure 4.23 - Converting the M:N Relationship into Two 1:M Relationships

Table name: STUDENT

Database name: Ch04_CollegeTry

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Table name: ENROLL

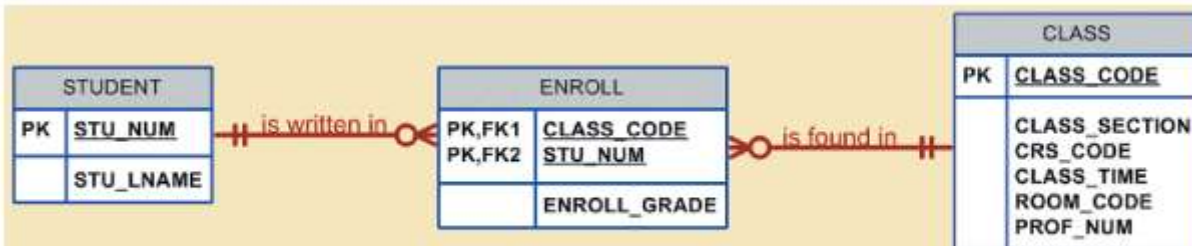
CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MMWF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MMWF 8:00-8:50 a.m.	KLR200	114

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Figure 4.25 - A Composite Entity in an ERD



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Putting together an ERD

Developing an ER Diagram

- Create a detailed narrative of the organization's description of operations
- Identify business rules based on the descriptions
- Identify main entities and relationships from the business rules
- Develop the initial ERD
- Identify the attributes and primary keys that adequately describe entities
- Revise and review ERD

Figure 4.26 - The First Tiny College ERD Segment

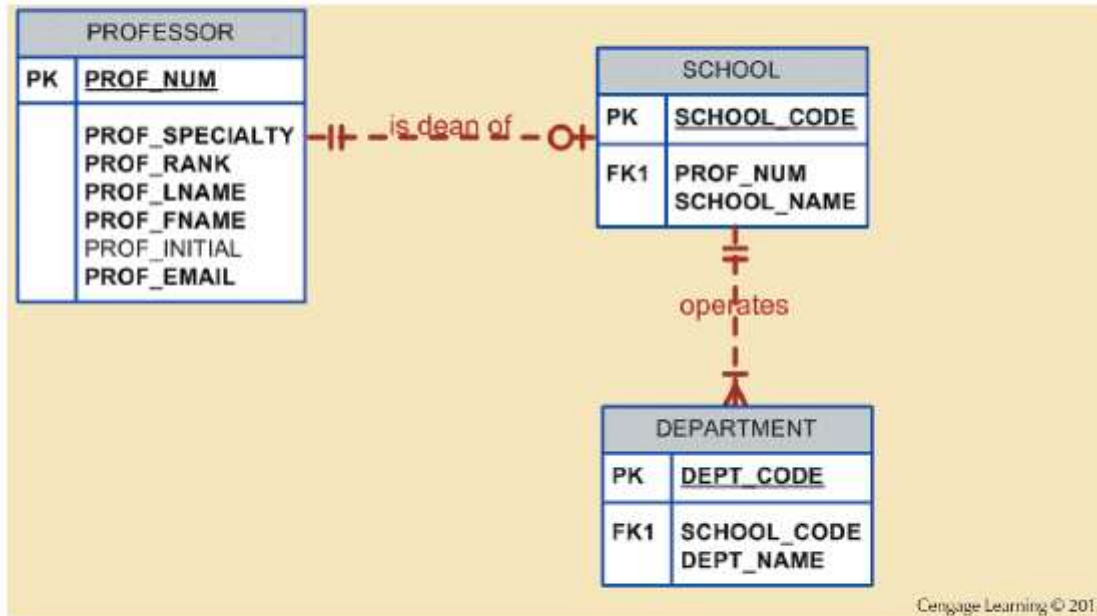


Figure 4.27 - The Second Tiny College
ERD Segment



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Figure 4.28 - The Third Tiny College ERD Segment

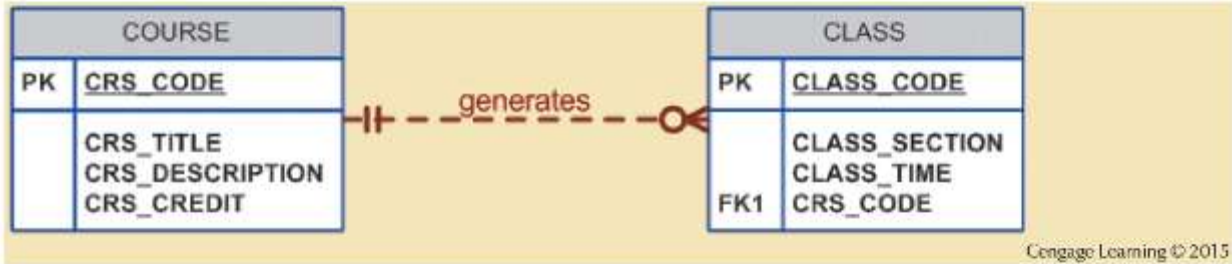


Figure 4.29 - The Fourth Tiny College ERD Segment

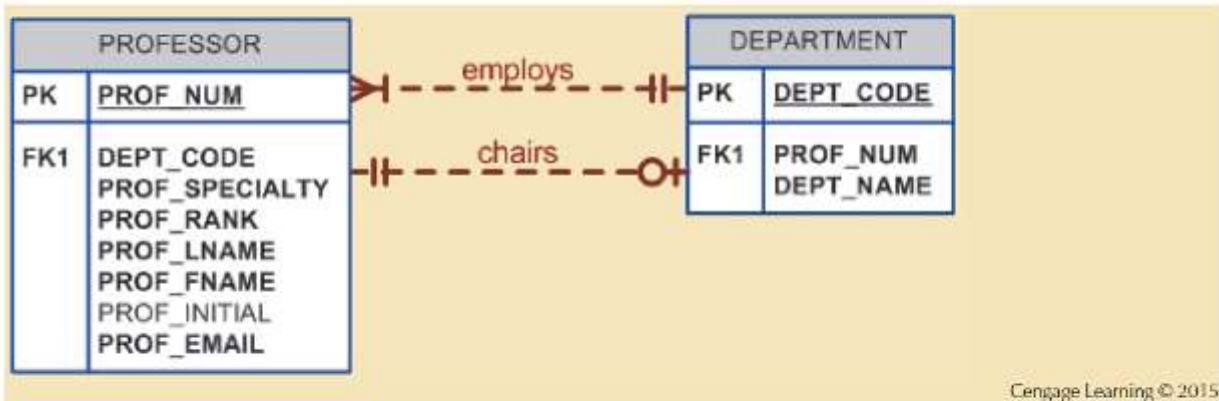


Figure 4.30 - The Fifth Tiny College ERD Segment

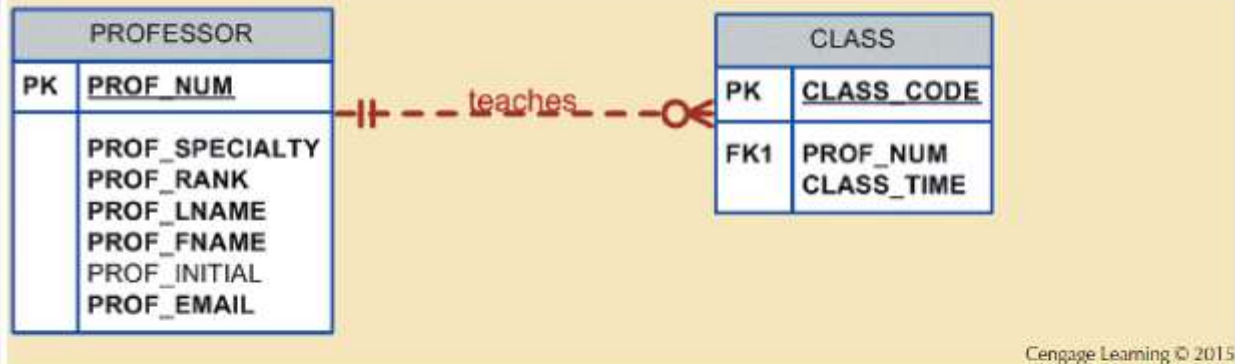


Figure 4.31 - The Sixth Tiny College ERD Segment



Figure 4.32 - The Seventh Tiny College
ERD Segment

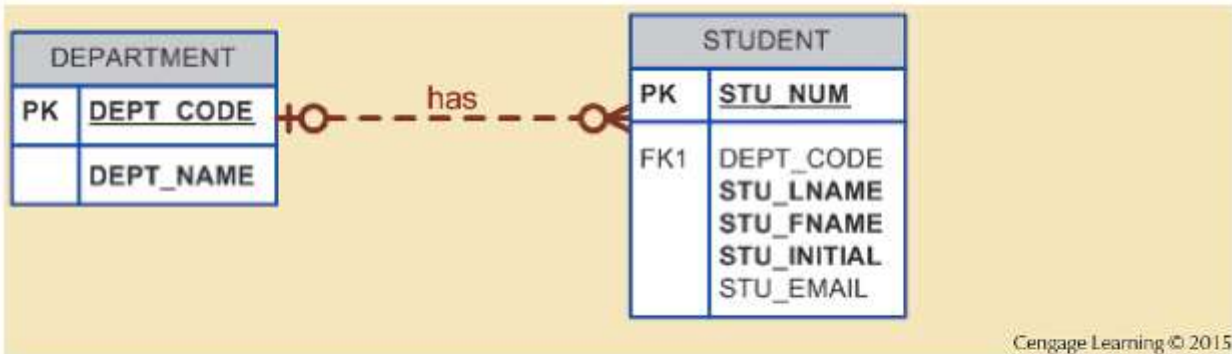
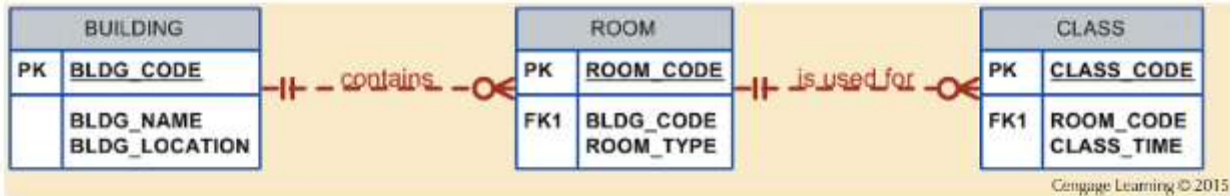


Figure 4.33 - The Eighth Tiny College ERD Segment



Figure 4.34 - The Ninth Tiny College ERD Segment



List of entities, relationships, connectivities

Table 4.4 - Components of the ERM

ENTITY	RELATIONSHIP	CONNECTIVITY	ENTITY
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	M:N	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS

Note: ENROLL is the composite entity that implements the M:N relationship "STUDENT enrolls in CLASS."

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The full schema

"All together now!"

